Lecture VII - NumPy for Scientific Computing

Programming with Python

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EP 7: Numbers Deck

The numbers deck is due

The investor wants a numbers deck. **Two weeks of orders** sit in the system, one row each: minutes, price, zone.

Tobi's plan is a **40-tab spreadsheet**, one tab per zone, copied by hand. He looks at the pile, then at his list-of-lists, and says the only true thing he'll say all week:

> **"We need a bigger boat than a list."**

Today we get the bigger boat: **NumPy**, one object that holds a thousand numbers and does arithmetic on all of them at once.

Warm-up

Question 1

Three questions from Episode 6. Commit. Hands up **before** the reveal.

```
from math import ceil

# which line works?
```

a) `ceil(3.2)` b) `math.ceil(3.2)` c) both

Answer 1

```
from math import ceil

print(ceil(3.2))
print(math.ceil(3.2))
```

```
4
```

```
NameError: name 'math' is not defined
[31mNameError[39m[31m:[39m name 'math' is not defined
```

a) `ceil(3.2)`: `from math import ceil` binds only the **name** `ceil`. The module `math` itself was never imported, so `math.ceil` has nothing to reach through.

Question 2

```
import random

random.seed(42)
print(random.randint(1, 20))
print(random.randint(1, 20))
print(random.randint(1, 20))
```

Tobi runs this exact script today and again tomorrow. Tomorrow's numbers are…

a) different: random is random b) an error: seed 42 was already used c) the same three numbers

Answer 2

```
import random

random.seed(42)   # today
print(random.randint(1, 20), random.randint(1, 20), random.randint(1, 20))
random.seed(42)   # tomorrow: a fresh run starts from the seed again
print(random.randint(1, 20), random.randint(1, 20), random.randint(1, 20))
```

```
4 1 9
4 1 9
```

c) the same three numbers. Every run starts from seed 42, so the stream replays from the top. That is the entire job of a seed: reproducible randomness. (Two batches *inside one run* would differ: the stream continues; a fresh run rewinds it.)

Question 3

```
import statistics

statistics.median([9, 2, 5])
```

returns…

a) `2` b) `5` c) an error: the list is not sorted

Answer 3

```
import statistics

statistics.median([9, 2, 5])
```

```
5
```

b) `5`: `median` sorts the values internally before picking the middle one. You never have to sort first; `5` is the middle of `2, 5, 9`.

A thousand at once

From a list to an array

A **NumPy array** holds many numbers under one name, like a list, but built for math. You make one from a list, then ask it about itself:

```
import numpy as np

prices = np.array([12.0, 9.0, 15.0])
print(prices)
print(prices.shape)   # how many, in each dimension
print(prices.dtype)   # what kind of number
print(prices.size)    # how many in total
```

```
[12.  9. 15.]
(3,)
float64
3
```

`np.array([...])` wraps a list; `np` is the nickname everyone uses. `.shape` is `(3,)`, `.dtype` is `float64`, `.size` is `3`. One `dtype` for the **whole** array: all elements share one type, part of what makes it fast.

One operation, every element

Here's the bigger boat. The deck needs **gross** prices: 19% VAT on all three. With a list you loop; with an array you just multiply:

```
# the painful way: a loop, item by item
gross = []
for p in [12.0, 9.0, 15.0]:
    gross.append(p * 1.19)
```

```
import numpy as np

prices = np.array([12.0, 9.0, 15.0])
print(prices * 1.19)   # every element, one expression
```

```
[14.28 10.71 17.85]
```

One operation lands on **all** elements at once: `[14.28 10.71 17.85]`. No loop, no `.append`, and on a thousand orders it's also far faster.

Your turn

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_07_a/



Any questions? Let me know!

Arrays from scratch

Two builders make evenly-spaced arrays without typing every number, handy for axes and ranges:

```
import numpy as np

print(np.arange(0, 10, 2))     # start, stop (excluded), step
print(np.linspace(0, 1, 5))    # start, stop (included), how many
```

```
[0 2 4 6 8]
[0.   0.25 0.5  0.75 1.  ]
```

`arange` walks by a **step** and stops before the end, just like `range`. `linspace` splits a span into a fixed **count** of points, endpoints included.

Predict: times two

Tobi multiplies a row of counts by two. What does this print?

```
print([1, 2, 3] * 2)
```

a) `[1, 2, 3, 1, 2, 3]` b) `[2, 4, 6]` c) an error

Predict. Pick a letter, then I reveal the answer.

Answer: repeat versus compute

```
import numpy as np

print([1, 2, 3] * 2)              # list → repeated
print(np.array([1, 2, 3]) * 2)    # array → doubled
```

```
[1, 2, 3, 1, 2, 3]
[2 4 6]
```

a) **`[1, 2, 3, 1, 2, 3]`**. That's a plain **list**, and `* 2` on a list *repeats* it. Wrap it in an array and the same `* 2` does the math instead.

Lists repeat; arrays compute. That's why we're here.

Predict: plus

Tobi adds this week's zone counts to last week's. What does this print?

```
print([1, 2, 3] + [10, 20, 30])
```

a) `[11, 22, 33]`, added pairwise b) an error: lists don't add c) `[1, 2, 3, 10, 20, 30]`

Predict. Pick a letter, then I reveal the answer.

Answer: lists glue, arrays add

```
import numpy as np

print([1, 2, 3] + [10, 20, 30])                        # list → glued
print(np.array([1, 2, 3]) + np.array([10, 20, 30]))    # array → added
```

```
[1, 2, 3, 10, 20, 30]
[11 22 33]
```

c) `[1, 2, 3, 10, 20, 30]`. On lists, `+` **glues** them end to end, exactly like `* 2` repeated. On arrays, `+` adds element by element, first with first, second with second.

Every operator follows the same rule: lists do list things, arrays do math.

Your turn

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_07_b/

Any questions? Let me know!

Ask the data

A comparison makes a mask

Compare an array to a number and you don't get one `True`/`False`. You get a **whole array** of them, one per element. That's a **mask**, and you can filter with it:

```
import numpy as np

times = np.array([25, 41, 18, 33])
print(times > 30)            # a True/False for every element
print(times[times > 30])     # keep only where the mask is True
```

```
[False  True False  True]
[41 33]
```

`times > 30` is the mask; `times[times > 30]` reads the array **through** the mask and returns just the matching values. No loop, no `if`.

Counting without a loop

A mask answers two investor questions at once. `.sum()` counts the `True`s (each counts as `1`); `.mean()` gives the **share** that are `True`:

```
import numpy as np

times = np.array([25, 41, 18, 33, 52, 29, 44, 12])
late = times > 30
print(int(late.sum()))              # how many were late
print(float(times[late].mean()))    # average of just the late ones
print(float(late.mean()))           # the SHARE that were late
```

```
4
42.5
0.5
```

`4` deliveries over 30 minutes, averaging `42.5`, and `late.mean()` says **half** the run was late, one line each, straight into the deck. `int(...)` and `float(...)` turn NumPy's numbers into plain Python numbers.

Predict: counting the Trues

What does calling `.sum()` on the mask give?

```
print((np.array([1, 5, 3]) > 2).sum())
```

a) `True` b) `2` c) `[False, True, True]`

Predict. Pick a letter, then I reveal the answer.

Answer: True counts as 1

```
import numpy as np

print(np.array([1, 5, 3]) > 2)            # [False  True  True]
print((np.array([1, 5, 3]) > 2).sum())    # Trues add up to 2
```

```
[False  True  True]
2
```

b) `2`: `.sum()` adds the mask up, and each `True` is worth `1`, each `False` `0`. Two elements clear the bar, so the count is `2`.

Summing a mask counts; averaging a mask shares. Same two tricks the lab asks for.

Predict: two conditions

The investor wants deliveries that took **more than 20 and less than 40** minutes. Tobi writes what he'd write for two numbers:

```
times = np.array([25, 41, 18, 33])
print(times > 20 and times < 40)
```

a) an error: ambiguous truth value b) `[ True False False  True]` c) `True`, both sides hold somewhere

Predict. Pick a letter, then I reveal the answer.

Answer: one `and`, four truths

```
import numpy as np

times = np.array([25, 41, 18, 33])
print(times > 20 and times < 40)
```

```
ValueError: The truth value of an array with more than one element is
ambiguous. Use a.any() or a.all()
[31mValueError[39m[31m:[39m The truth value of an array with more than
one element is ambiguous. Use a.any() or a.all()
```

a) an error. `and` asks the left side "are you true?", and a four-element mask has no single answer. Between two masks use `&`, and wrap each side in parentheses (`&` binds tighter than `>`):

```
print((times > 20) & (times < 40))    # element by element: & combines masks
```

```
[ True False False  True]
```

Your turn

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_07_c/



Any questions? Let me know!

The days-by-zones grid

A grid of numbers

Real data isn't one row. Stack rows and you get a **2D array**: here three days (rows) across four zones (columns: Nord, Sued, Hafen, Altstadt):

```
import numpy as np

deliveries = np.array([[ 9, 14, 11,  6],
                       [15, 12,  8,  9],
                       [13, 20, 16, 11]])
print(deliveries.shape)      # (rows, columns) → (3, 4)
print(deliveries[0, 2])      # row 0, column 2
```

```
(3, 4)
11
```

`.shape` is now `(3, 4)`: three days, four zones. One index picks the **row**, a second the **column**: `deliveries[0, 2]` is day 0, zone 2 (Hafen).

Predict: the shape

Tobi builds a grid with two more days. What does `.shape` say?

```
grid = np.array([[ 9, 14, 11,  6],
                 [15, 12,  8,  9],
                 [13, 20, 16, 11],
```

```
                [10, 17, 12,  8],
                [ 7, 11,  9,  5]])
print(grid.shape)
```

a) `(20,)` b) `(5, 4)` c) `(4, 5)`

Predict. Pick a letter, then I reveal the answer.

Answer: rows first, then columns

```
import numpy as np

grid = np.array([[ 9, 14, 11,  6],
                 [15, 12,  8,  9],
                 [13, 20, 16, 11],
                 [10, 17, 12,  8],
                 [ 7, 11,  9,  5]])
print(grid.shape)   # (rows, columns)
print(grid.size)    # rows × columns
```

```
(5, 4)
20
```

b) `(5, 4)`. `.shape` is always `(rows, columns)`: five inner lists make five rows, each with four numbers. `(20,)` would be one flat row of twenty; `20` is `.size`, the total count.

Same order as indexing: `grid[row, column]`, rows first.

Which way to collapse?

To sum a grid you must say **which way** to collapse it. The `axis` tells NumPy which direction disappears:

```
        Nord  Sued  Hafen  Altstadt
day0  [  9    14    11      6  ]
day1  [ 15    12     8      9  ]
day2  [ 13    20    16     11  ]

axis=0 collapses DOWN the rows:
         ↓     ↓     ↓      ↓
        37    46    35     26      one number per column (zone)

axis=1 collapses ACROSS the columns:
day0 → 40   ·   day1 → 44   ·   day2 → 60      one number per row (day)
```

`axis` in code

The same grid, the same two collapses, one keyword each:

```
import numpy as np

deliveries = np.array([[ 9, 14, 11,  6],
                       [15, 12,  8,  9],
                       [13, 20, 16, 11]])
print(deliveries.sum(axis=0))   # DOWN the rows → per zone
print(deliveries.sum(axis=1))   # ACROSS the columns → per day
```

```
[37 46 35 26]
[40 44 60]
```

`axis=0` collapses DOWN the rows, one number per column (zone): `[37 46 35 26]`. `axis=1` collapses across, one per day: `[40 44 60]`.

Your turn

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_07_d/

Any questions? Let me know!

Where does the max sit?

The per-zone totals answer "how many?", but the investor asks "**which** zone?". `argmax` tells you **WHERE** the maximum sits, as an index:

```
import numpy as np
```

```
deliveries = np.array([[ 9, 14, 11,  6],
                       [15, 12,  8,  9],
                       [13, 20, 16, 11]])
zone_totals = deliveries.sum(axis=0)   # [37 46 35 26]
print(int(zone_totals.argmax()))        # index of the biggest
```

```
1
```

`max` would give the value `46`; `argmax` gives its **position**, `1`. The busiest zone is **Sued**, sitting at index 1, not at the front. Position, not value: that's the whole point of `argmax`.

Predict: which axis is which?

The investor wants **four** numbers, one per zone. Which call?

```
week = np.array([[ 9, 14, 11,  6],
                 [15, 12,  8,  9],
                 [13, 20, 16, 11]])
```

a) `week.sum(axis=1)` b) `week.sum()` c) `week.sum(axis=0)`

Predict. Pick a letter, then I reveal the answer.

Answer: days collapse, zones stay

```
import numpy as np

week = np.array([[ 9, 14, 11,  6],
                 [15, 12,  8,  9],
                 [13, 20, 16, 11]])
print(week.sum(axis=0))
```

```
[37 46 35 26]
```

c) **`week.sum(axis=0)`**. Four zones means four numbers, so the **days** must disappear: `axis=0` collapses DOWN the rows, one number per column (zone).

a) would give **three** numbers (one per day); b) would give **one** number: the grand total, `144`.

Your turn

Open the exercise

Scan the QR or type the link:

python.tobiasvlcek.com/notebooks/ex_07_e/



Any questions? Let me know!

Wrap-up

Three things to remember

1. **One array, one operation.** `np.array([...])` holds many numbers; arithmetic hits **every element at once**. No loop. Ask it `.shape`, `.dtype`, `.size` to know what you're holding.
2. **A comparison makes a mask.** `arr > 30` is a True/False array: `arr[mask]` filters, `.sum()` counts the Trues, `.mean()` gives their share.
3. **In 2D, pick an axis.** `axis=0` collapses DOWN the rows (one number per column), `axis=1` across the columns (one per row); `argmax` tells you where the maximum sits.

Next episode

> **Note**
> **Next episode** starts with **Checkpoint 4**: 40 minutes, AI allowed, everything from Episodes 6-7. Then the investor opens a data room, and Tobi lets an AI write his pandas: the arrays get column names, and eighty orders become a table you can query.

Leaving the Browser

From a browser tab to your machine

For seven sessions everything ran in a browser tab. From Session VIII the labs run on **your own laptop**, with three free tools:

- **uv** installs Python and runs your scripts
- **Zed** is the editor you write code in
- **Mistral Vibe** is the AI agent inside Zed

We install all three **now, together**. Today's lab starts for you once your setup passes the check.

Step 1: uv and your course folder

Follow the uv guide from the top through *Adding packages*, one command at a time. It does three things:

- installs **uv** and **Python**
- creates the folder `python-labs`, where every lab from Session VIII lives
- adds the two packages those labs need: pandas and matplotlib

> ! **Important**
> **Type each command once.** A second `uv init` builds a second `python-labs` inside the first, and uv then refuses to work in either.

Step 2: Zed and Mistral Vibe

Follow *AI inside Zed* on the AI Tools page, steps 1 to 5:

1. Install **Zed** from zed.dev and open it once
2. In a terminal: `uv tool install mistral-vibe`
3. Run `vibe` and sign in with your Mistral account from last session
4. In Zed's command palette (`Cmd/Ctrl+Shift+P`): **`zed: acp registry`**, then install **Mistral Vibe**
5. **File → Open Folder** and pick `python-labs`

> **Tip**
> **`vibe` not found?** Run `uv tool update-shell`, then close and reopen the terminal.

The check before the lab

In Zed, with `python-labs` open, open the terminal (`Ctrl+`` ` ``, or **`terminal panel: toggle`** in the command palette) and run:

```
uv run python -c "import pandas, matplotlib; print('ready')"
```

- The terminal prints `ready`
- In the agent panel (the sparkle icon, bottom right), pick **Mistral Vibe** and ask: *What does `main.py` do?* It answers

Show me both, then open the lab.

> ! **Important**
> **Stuck?** Raise your hand early. No laptop that runs Zed, or only an iPad? Tell me. You start the **Blank** template at github.com/codespaces (a Linux computer in a browser tab), do Step 1 in its terminal, and show me `ready` there.

To the Lab

After the check: the lab

- Head to the lab notebook: Episode 7: The Numbers Deck
- You'll count and average the critical deliveries with masks, fix two of Tobi's numbers, and find the winning zone in the days-by-zones grid
- AI is allowed, so try the course chatbot or your Mistral account
- It still runs in your browser; unfinished parts are homework before **Checkpoint 4**

> ! **Important**
> **Download your `.py` before you leave.** Closing the tab without downloading loses your work. **Next session, bring this laptop:** from Episode 8 the labs run in Zed.

Literature

Books to start with

- Downey, A. B. (2024). Think Python: How to think like a computer scientist (Third edition). O'Reilly. Link to free online version
- Elter, S. (2021). Schrödinger programmiert Python: Das etwas andere Fachbuch (1. Auflage). Rheinwerk Verlag.

> i **Note**
> NumPy has excellent free docs: the NumPy absolute beginner's guide covers everything in this session and a little more.

For more, see the literature list of this course.